

$$Power = \epsilon \sigma A T^4$$

Soru 1

Stefan-Boltzmann yasası $I_T = \sigma T^4$

$$T \rightarrow 2T \text{ olursa } \sigma (2T)^4 = 16\sigma T^4 =$$

Yani yayılan güç 16 kat artar $16 I$

Soru 2

$$\lambda_{max} T = 2898 \times 10^{-3} \text{ m.k}$$

a) Formülde k yerine 5800 K (güneş ısı) değeri yazalım.

$$\lambda_{max} = \frac{2898 \times 10^{-3}}{5800} = \boxed{4.996 \times 10^{-7} \text{ m}}$$

b) λ_{max} değeri elektromanyetik spektruma göre Görünür ışık bölgesindedir.

Soru 3

$$\lambda_{max} T = 2898 \times 10^{-3} \text{ m.k denklemini kullanacağız}$$

$$40 \cdot 10^{-6} T = \frac{2898 \times 10^{-3}}{40 \cdot 10^{-6}} = \frac{2898}{40} = T$$

$$\boxed{T = 72.45 \text{ kelvin}}$$

Planck sabiti = $6,63 \times 10^{-34}$ Joule

SORU 4) $E = h\nu$ formülüne

a) $1 \text{ MHz} = 1 \times 10^6 \text{ Hz}$ $E = (6,63 \times 10^{-34}) (1 \times 10^6)$

$\frac{6,63 \times 10^{-28} \text{ Joule}}{1,6 \times 10^{-19} \text{ e}^-} = \boxed{4,14 \times 10^{-9} \text{ eV}}$

(Coulomb sabiti ne buluyoruz)

b) $E = 6,63 \times 10^{-34} \times 5 \cdot 10^{14} = 33,15 \times 10^{-20} \text{ Joule}$

$\frac{33,15 \times 10^{-20}}{1,6 \times 10^{-19}} = \boxed{2,071875 \text{ eV}}$

c) $E = 6,63 \times 10^{-34} \times 3 \times 10^{18} = 19,89 \times 10^{-16} \text{ Joule}$

$\frac{19,89 \times 10^{-16} \text{ Joule}}{1,6 \times 10^{-19}} = 12431,25 \text{ eV}$

SORU 5) $\phi = 1,5 \text{ eV}$, $\lambda = 3000 \text{ Å}$

$m_{\text{electron}} = 9,1 \times 10^{-31} \text{ kg}$ $3000 \times 10^{-10} \text{ m}$

$c = 3 \times 10^8 \text{ m/s}$

a) $E = \frac{hc}{\lambda} = \frac{6,63 \times 10^{-34} \times 3 \times 10^8}{3000 \times 10^{-10}} = 6,63 \times 10^{-19} \text{ eV}$

$E_{\text{kinet}} = h\nu - \phi$

$4,14 - 1,5 = 2,64 \text{ eV}$

$\frac{6,63 \times 10^{-19}}{1,6 \times 10^{-19}} = 4,14 \text{ eV}$

yaklaşık 4,14 eV

$\boxed{2,64 \text{ Volt}}$

$$b) k_{\max} = \frac{1}{2}mv^2$$

$$2,64 \text{ eV} = 2,64 \times 1,6 \times 10^{-19} \text{ joule} = 4,22 \times 10^{-19} \text{ joule}$$

$$v^2 = \sqrt{\frac{4,22 \times 10^{-19}}{9,11 \times 10^{-31}}} \approx 9,61 \times 10^5 \text{ m/s}$$